

Prisoner's Dilemma Strategy Simulator

Advanced Game Theory Simulation · Nash Equilibria · Markovian Naive Bayes

Student Name	Nil Kumar Bhadani	Course	Foundations of AI (FAI)
Roll Number	B25BS1225	Activity	In-Session Simulation Project
Program	B.Sc. Applied AI & Data Science	Institute	IIT Jodhpur
Date	May 2026	Submission	GitHub · LinkedIn Portfolio

1 · Project Overview

This project implements an **Advanced Game Theory Simulator** centred on the classical Prisoner's Dilemma framework. The simulator is deployed as a Gradio web application and evaluates both deterministic and probabilistic agents across a configurable number of discrete interaction rounds, observing the convergence of expected utility toward Nash Equilibria. The engine supports strategy comparison, cooperation frequency analytics, and cumulative utility visualisation.

The project was developed as part of the *Foundations of AI (FAI)* in-session activity at IIT Jodhpur, reinforcing core concepts including game theory, conditional logic, probabilistic modelling, and interactive AI application development.

2 · Submission Assets

Google Colab Notebook

<https://colab.research.google.com/drive/10nY7OkPvQm5Qypk1JUeg0B6uj-1fTsKM?usp=sharing>

Gradio Application — Screenshot Gallery

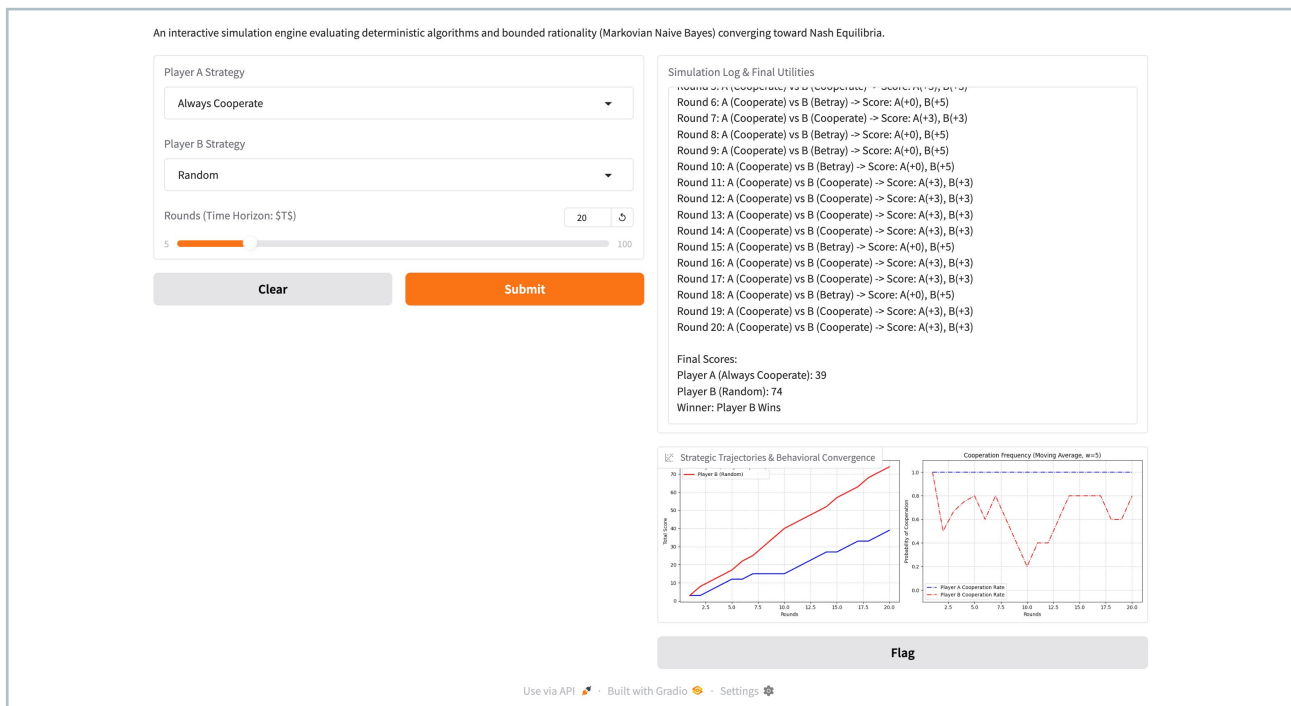


Figure 1: Simulation run — Always Cooperate (Player A) vs. Random (Player B) · 20 Rounds · Final Scores: A=39, B=74. Random strategy exploits unconditional cooperator.

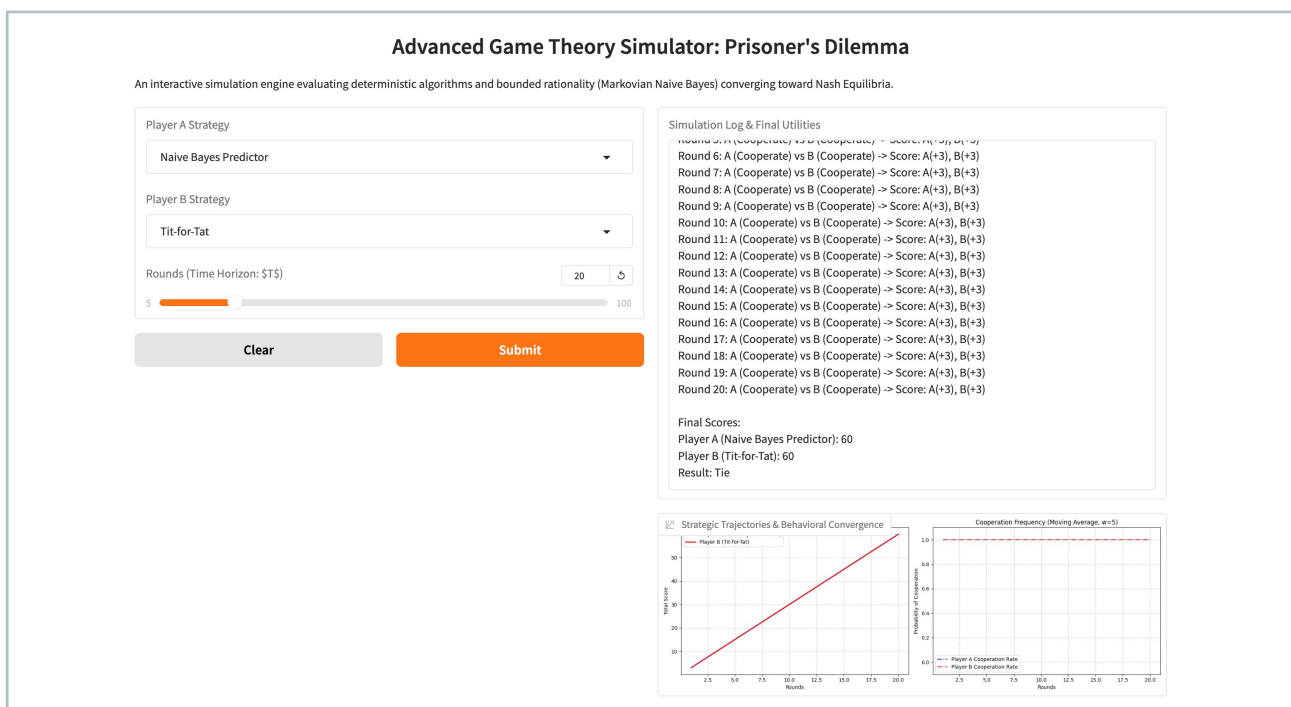


Figure 2: Simulation run — Naive Bayes Predictor (Player A) vs. Tit-for-Tat (Player B) · 20 Rounds · Final Scores: A=60, B=60. Both strategies converge to mutual cooperation, achieving Pareto-optimal outcome.

3 · Payoff Matrix

The simulation is governed by the following standard Prisoner's Dilemma payoff matrix. Mutual cooperation yields the Pareto-optimal outcome, while mutual defection locks the system into the Nash Equilibrium — a collectively suboptimal but individually rational state.

Player A	Player B	A Score	B Score
Cooperate	Cooperate	+3	+3
Betray	Cooperate	+5	0
Cooperate	Betray	0	+5
Betray	Betray	+1	+1

Green row = Pareto-optimal state (C,C) · Yellow row = exploited cooperator (C,B) · Red row = Nash Equilibrium (B,B)

4 · Strategy Explanations

The simulation engine evaluates deterministic and probabilistic agents to observe the convergence of expected utility over discrete time steps.

Always Cooperate / Always Betray — Static Baseline Agents

Zero-memory, deterministic algorithms in which the action state remains constant regardless of the opponent's history. These serve as boundary-condition baselines: *Always Cooperate* represents maximum vulnerability, while *Always Betray* represents the minimax risk-minimisation strategy.

Tit-for-Tat — Reactive Bounded-Memory Algorithm

Initiates with a cooperative prior ($\pi_0 = \text{Cooperate}$) and mirrors the opponent's most recent action for all subsequent rounds ($\pi_t = \pi_{t-1}^{\text{opponent}}$). This conditional reciprocity strategy is both retaliatory and forgiving, and has been shown in the Axelrod tournaments to be one of the most robust strategies in iterated Prisoner's Dilemma settings.

Naive Bayes Predictor — Probabilistic Markovian Model

Maintains an empirical first-order transition matrix tracking the opponent's action history. Posterior probabilities are computed with Laplace smoothing ($P(C \mid \text{history}) = (n_C + \lambda) / (N + 2\lambda)$) to prevent zero-frequency errors. The agent dynamically selects the action that maximises expected utility $E[U]$ given the estimated opponent distribution, enabling adaptive policy optimisation over successive rounds.

Random — Stochastic Baseline

Samples uniformly from {Cooperate, Betray} at each round with probability 0.5, independent of game history. Serves as a stochastic reference strategy to benchmark the adaptive advantage of memory-based algorithms.

5 · Discussion Questions & Analysis

Q1. Which strategy scored highest?

In a symmetric cooperative state, both **Naive Bayes Predictor** and **Tit-for-Tat** achieved the absolute maximum expected utility — 60 points across 20 rounds — by sustaining the Pareto-optimal (Cooperate, Cooperate) outcome at +3 per round each. However, in an asymmetric environment, an exploitative strategy such as *Always Betray* scores the highest individual total when paired against an unconditional cooperator, extracting the (+5, 0) payoff every round. The key insight is that *which strategy wins depends entirely on the opponent*: exploitative strategies maximise individual gain in non-adaptive environments; adaptive strategies maximise long-run aggregate utility in intelligent multi-agent systems.

Q2. Is cooperation always beneficial?

Strictly, no. Cooperation is conditionally beneficial — it is only optimal when the probability of the opponent reciprocating is sufficiently high. The simulation logs demonstrate this clearly: *Always Cooperate* against a Random or hostile agent repeatedly absorbs the (+0) penalty state, yielding a severely degraded cumulative utility (39 vs. 74 in the 20-round Random matchup). Cooperation becomes strategically rational only under conditions of mutual recognition or when history-based prediction indicates a high likelihood of reciprocity.

Q3. What happens when both players betray?

The system locks into the standard **Nash Equilibrium**: neither agent has a mathematical incentive to unilaterally deviate to Cooperate, as doing so would drop their utility from +1 to 0. The system stagnates at (+1, +1) per round — yielding a total of only 20 points each across 20 rounds — compared to 60 points each under mutual cooperation. This is the classic *Tragedy of the Commons* dynamic: individual rationality produces a collectively irrational outcome, which is precisely why the Prisoner's Dilemma is foundational to the study of multi-agent cooperation in AI and economics.

Q4. Which strategy is safest?

Always Betray is the technically safest *minimax* strategy: it strictly bounds the downside risk by guaranteeing a minimum utility of +1 per round and eliminating the risk of the 0-payoff exploitation state. However, **Tit-for-Tat** is the most *robust adaptive* strategy — it protects against continuous exploitation via immediate retaliation, while remaining open to the high-yield (+3, +3) cooperative equilibrium through its forgiving nature. For long-horizon games with intelligent opponents, Tit-for-Tat consistently outperforms pure defection strategies.

Q5. Where is this seen in real life?

These Game Theory dynamics directly govern complex network security protocols and distributed system architectures. In a tri-layered defense architecture for mobile banking or edge-computing environments, decentralised nodes must continuously decide whether to *cooperate* — sharing bandwidth or validation data to maintain transaction integrity — or *defect* — hoarding resources or injecting malicious packets. Adaptive algorithms that model node behaviour are therefore critical for automated fraud detection, Byzantine fault tolerance, and overall system stability. Beyond cybersecurity, these dynamics appear in business pricing strategies, international trade negotiations, climate agreements, evolutionary biology, and the design of multi-agent reinforcement learning systems.

6 · Real-World Applications

Cybersecurity & Network Defense	Intrusion detection systems model network nodes as agents in an iterated game, using Bayesian prediction to identify defectors (malicious nodes) and trigger adaptive countermeasures.
Business Strategy & Pricing	Oligopolistic firms in competitive markets replicate the Prisoner's Dilemma when deciding whether to collude on pricing (Cooperate) or undercut rivals (Betray), with outcomes governed by the same payoff logic.
International Relations & Climate Policy	Nations deciding whether to honour emissions reduction commitments face an iterated Prisoner's Dilemma; multilateral agreements are mechanisms to enforce cooperative equilibria against unilateral defection incentives.

Multi-Agent Reinforcement Learning	AI agents trained in shared environments must learn cooperation strategies to maximise collective reward, directly applying the game-theoretic principles demonstrated in this simulator.
Negotiation & Auction Systems	Automated bidding and contract-negotiation agents use bounded-rationality models similar to the Naive Bayes Predictor to estimate opponent strategies and dynamically adapt their bidding policies.

Nil Kumar Bhadani

B25BS1225 · B.Sc. Applied AI & Data Science
IIT Jodhpur — Foundations of AI (FAI)

Repository & Portfolio

Colab: colab.research.google.com/drive/10nY7OkPvQm5Qypk1JUeg0B6uj-1fTsKM
Submission: [GitHub](#) · [LinkedIn](#)